Sicher C Compiler Proof of Concept

Technology Stack
- Python3
- tkinter
- GNU/Linux/POSIX
- x86-64 ELF Linux

Usage
- Ensure one has access to a c source file in a known path
- Ensure sicher.py can take args
- ./sicher.py [c source file] [visualisation args]

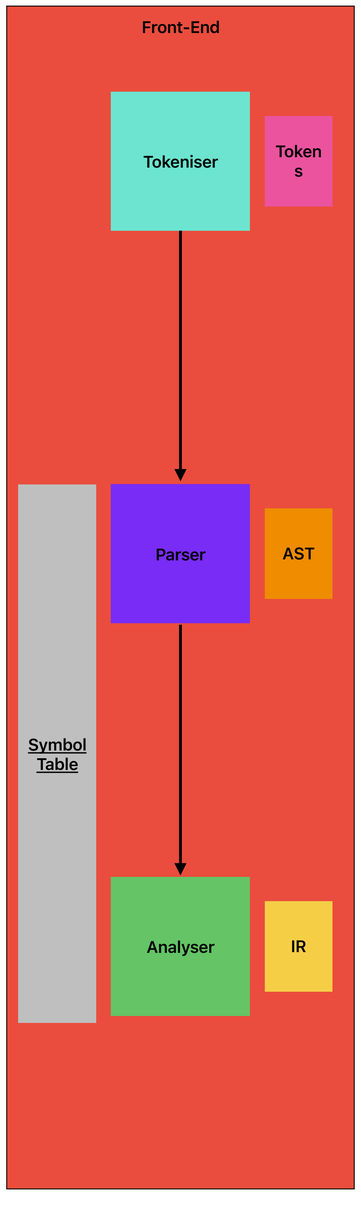
Abstract
- Target a simple dangling pointer source file
- Namely dangling.c
- Lex into tokens
- Parse into an AST
- Generate IR from AST
- Do compile time optimisations/ms countermeasures on AST
- Code Generate and do runtime code optimisations and ms countermeasures
- Log countermeasures/optimisations
- Use TKinter front end to allow for opening/compiling a given source file then show all the optimisations/ countermeasures from the text file in stack like structures
- **Do this all in tkinter for the pic**

Diff from PoC to Release
- Sicher will be written in C? Unlike poc's Python
- Could still use tkinter but should probably use OpenGL/ SDL2 or raylib et cetera.
- Will try to match c23 when possible
- May have larger scope such as C pre-processor, allow for more of the standard lib and so on
- Will be a lot harder to implement! But will be more performant and bespoke!

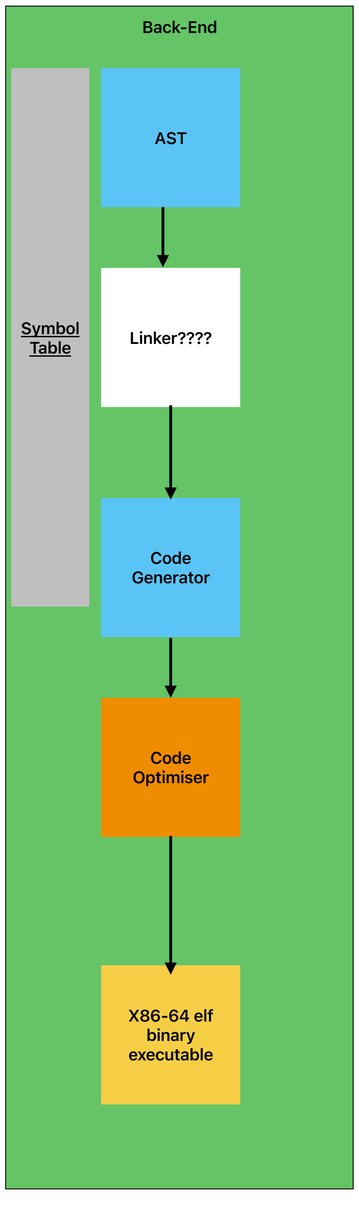
Memory Safe Features

Code Optimisations

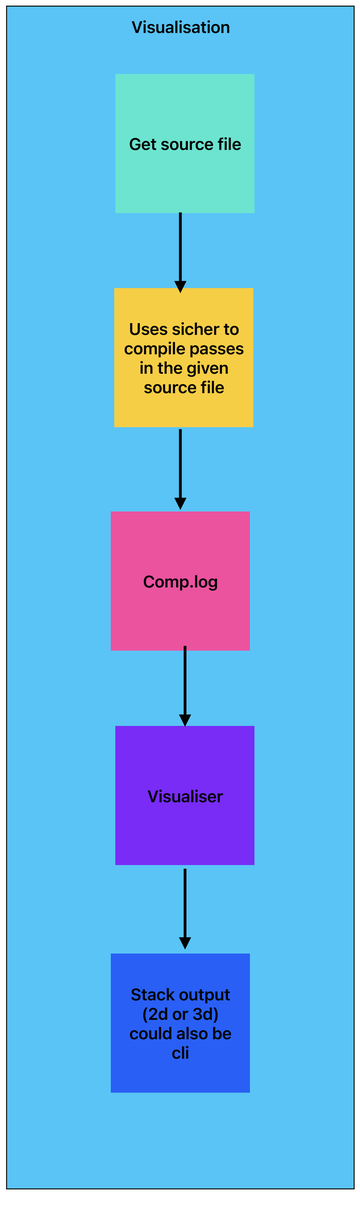
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MemorySafe C Compiler

